Generative AI Foundations for Engineers

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CLAIR SULLIVAN
& ASSOCIATES

Schedule

9:00 - 9:30	Check in, breakfast
9:30 - 9:45	Welcome and opening remarks
9:45 - 10:00	Overview of event
10:00 - 11:45	Module 1
12:00 - 12:30	WORKING LUNCH
12:30 - 2:00	Module 2
2:00 - 2:15	BREAK
2:15 - 3:45	Module 3
3:45 - 4:00	BREAK
4:00 - 4:20	Close competition
4:20 - 4:30	Announce winners, prizes!
4:30 - 5:30	Celebration and networking

Introduction

Agenda

- Brief introduction to the people in the room
- Structure for today
  - *Brief* lectures
  - Activities
  - Challenges (what is scored!)
- Module 1: Generative AI Foundations & LangChain Basics
- Module 2: Retrieval-Augmented Generation (RAG) & External Data Integration
- Module 3: Agents and Tool Use
- Wrap up, announcement of winners!

A brief survey:

PollEv.com/clairsullivan399

Module 1:
Generative AI Foundations & LangChain Basics

What is Generative AI (GenAI)?

- Traditional AI (AKA machine learning) focuses on using math to make predictions
- GenAI **creates** new content
- Trained on large datasets using deep learning to learn patterns and generate novel output
- Examples of GenAI in action

How Large Language Models (LLMs) Work

- Transformer models: the backbone of LLMs
  - Captures relationships between words across long texts
  - Key innovation in models like GPT, BERT, etc.
- Text is broken into **tokens**
- LLMs have a maximum number of tokens they can process: **context window**
  - When exceeded, older tokens are forgotten ("sliding window effect")
  - Varies by model
  - Longer context ≠ perfect memory
- LLMs are asked to do things through their **prompts**
- Limitations of LLMs
  - **Hallucinations**
  - Training data limitations (date, subject matter)
  - Computation cost

Introduction to LangChain

- Framework (Python, JS) for building applications powered by LLMs
- Why use it?
  - Open source
  - Simplifies development
  - Has wrappers for most common LLMs, cloud platforms
  - Manages context
  - Integrates with many data types, APIs
  - Chains, agents, tools easily created
- Cons of using LangChain
  - Debugging issues harder compared to direct API calls
  - Latency
  - In *active* state of development
  - LangGraph migration

Introduction to AWS Bedrock

- Fully managed service that provides foundation models via APIs
  - Anthropic, Meta, Mistral, Amazon Titan
- Enables creation of GenAI applications without managed infrastructure
- Seamless AWS integration with tools like S3, SageMaker, etc.
  - Including security features like IAM and VPC
- Built-in knowledge bases for retrieval-augmented generation (RAG)
- Integrated guardrails that allow you to configure content filters, blocklists, and moderation policies directly on the model outputs

What we CAN do today

- Introduce all of the basic concepts that you need in order to start creating GenAI applications
- Keep it simple
- Basic use patterns of GenAI, LangChain, Bedrock
- Make some things that work!

What we won’t be able to cover today

- Going beyond the basics
- Discussing every different approach to coding a problem
  - Includes all of the different ways to do a thing with LangChain and its ever-evolving suite of tools
- Integrate with a ton of different AWS products
- Use PII, real Bill data
- Discuss productionalizing GenAI, AIOps

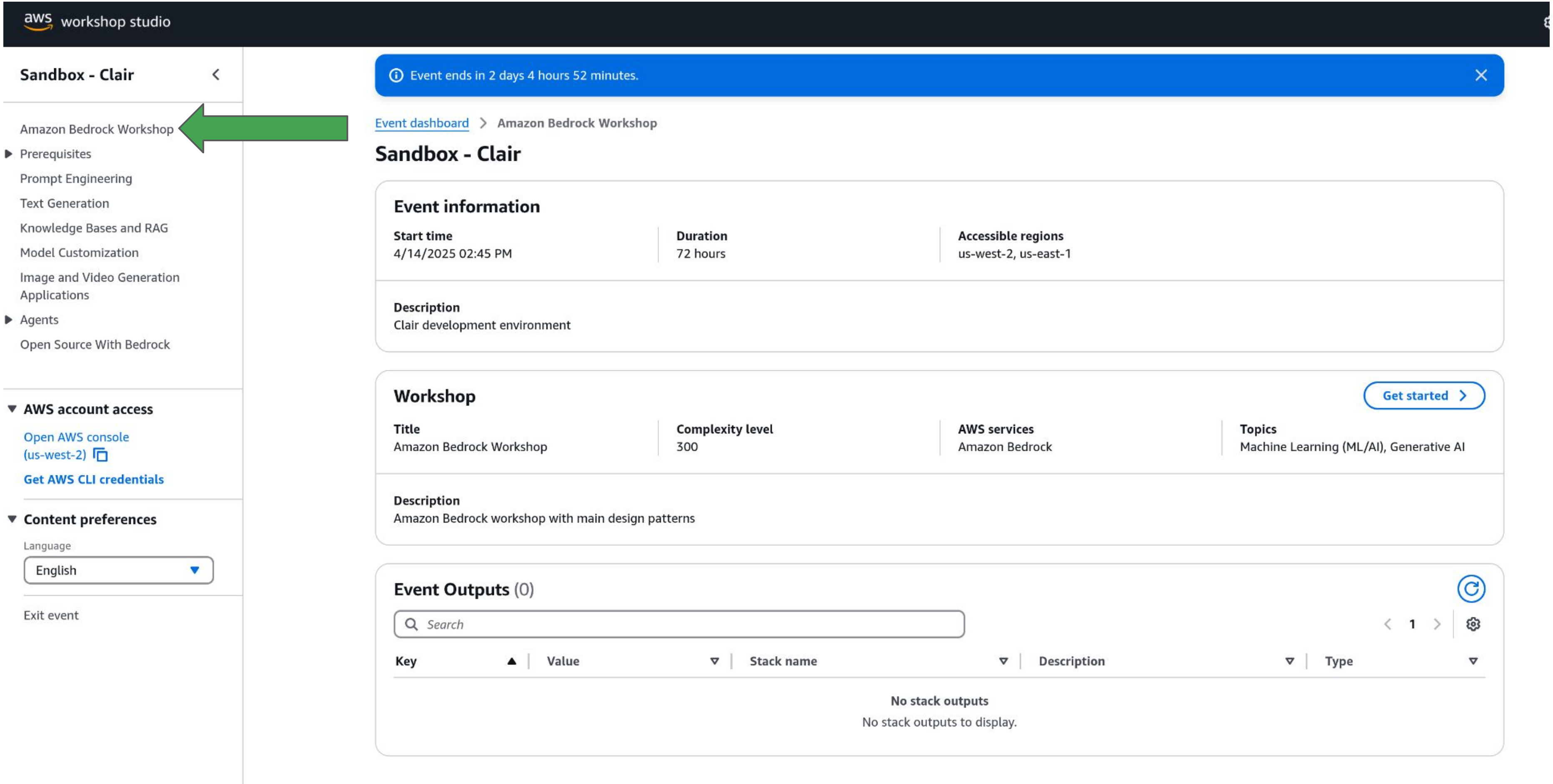
Structure for the hands on work

- Three topic areas
  - Code generation
  - Structured data analysis
  - Unstructured data analysis
- You don’t have to stick with one topic area or submit solutions for every question
- Start by going through the “Activity” notebooks
- You will be scored based on the “Challenges”
- You are *strongly* encouraged to work as a team, plan out the work, “divide and conquer”

Ground Rules

- **Have fun!!!**
- Experiment
- You will get deprecation warnings and that is OK
- Engage with everyone on your team
- Extra points can be awarded at any time for great *shared* learnings
- There are many prizes up for grabs, not just high scores
- *The Honor System: many of the challenges can be solved to some extent with traditional programming. For the answers you submit, please be honor-bound to submit your answers created via a GenAI solution.*
- *The Honor System II: Resist the urge to copy and paste your team's solutions. Try to create your own answer, which will likely be a different result than your teammates.*

How to get started with AWS Workshop


aws workshop studio
Sandbox - Clair
Amazon Bedrock Workshop
Prerequisites
Prompt Engineering
Text Generation
Knowledge Bases and RAG
Model Customization
Image and Video Generation Applications
Agents
Open Source With Bedrock
AWS account access
Open AWS console (us-west-2)
Get AWS CLI credentials
Content preferences
Language
English
Exit event
Event ends in 2 days 4 hours 52 minutes.
Event dashboard
Amazon Bedrock Workshop
Sandbox - Clair
Event information
Start time
4/14/2025 02:45 PM
Duration
72 hours
Accessible regions
us-west-2, us-east-1
Description
Clair development environment
Workshop
Get started
Title
Amazon Bedrock Workshop
Complexity level
300
AWS services
Amazon Bedrock
Topics
Machine Learning (ML/AI), Generative AI
Description
Amazon Bedrock workshop with main design patterns
Event Outputs (0)
Search
1
Key
Value
Stack name
Description
Type
No stack outputs
No stack outputs to display.

Event ends in 2 days 4 hours 50 minutes.

Amazon Bedrock Workshop

Welcome to "Amazon Bedrock" workshop!

The goal of this workshop is to give you hands-on experience in leveraging foundation models (FMs) through Amazon Bedrock. Amazon Bedrock is a fully managed service that provides access to FMs from third-party providers and Amazon; available via an API. With Bedrock, you can choose from a variety of models to find the one that's best suited for your use case.

Within this series of labs, you will go through some of the most common Generative AI usage patterns we are seeing with our customers. You will explore techniques for generating text and images, and learn how to improve productivity by using foundation models to help in composing emails, summarizing text, answering questions, building chatbots, creating images, and generating code. You will gain hands-on experience using Bedrock APIs, SDKs, and open-source software, such as LangChain and FAISS, to implement these usage patterns.

This workshop is intended for developers and solution builders. For example:

- Financial, Data and Business Analysts
- Generative AI developers
- C-suite executives: Understand how guardrails enhance data security and compliance.
- Marketing and sales teams: Learn how guardrails maintain brand voice and customer engagement.
- IT and DevOps teams: Explore the technical implementation of guardrails.
- Legal teams: Delve into how guardrails can help in compliance with laws and regulations.
- Data scientists, Machine Learning engineers, Developers, Product managers: Dive into the technical details and implementation of guardrails.
- Security & compliance teams: Understand how guardrails enhance data protection and compliance.

What's included in this workshop:

- Prompt Engineering [Estimated time to complete - **15 mins**]
- Text Generation [Estimated time to complete - **25 mins**]
- KnowledgeBase and RAG [Estimated time to complete - **30 mins**]
- Model Customization [Estimated time to complete - **35 mins**] - **Note** - you will not be able to run this section on AWS workshop studio.
- Images and Multimodal [Estimated time to complete - **35 mins**]

Prerequisites

Overview

This hands-on workshop, aimed at developers and solution builders, introduces how to leverage foundation models (FMs) through Amazon Bedrock. This code goes alongside the self-paced or instructor lead workshop here - https://catalog.us-east-1.prod.workshops.aws/amazon-bedrock/en-US

Please follow the prerequisites listed in the link above or ask your AWS workshop instructor how to get started.

Amazon Bedrock is a fully managed service that provides access to FMs from third-party providers and Amazon; available via an API. With Bedrock, you can choose from a variety of models to find the one that's best suited for your use case.

Within this series of labs, you'll explore some of the most common usage patterns we are seeing with our customers for Generative AI. We will show techniques for generating text and images, creating value for organizations by improving productivity. This is achieved by leveraging foundation models to help in composing emails, summarizing text, answering questions, building chatbots, and creating images. While the focus of this workshop is for you to gain hands-on experience implementing these patterns via Bedrock APIs and SDKs, you will also have the option of exploring integrations with open-source packages like LangChain and FAISS.

Note that you have two Options to run this workshop:

- At an AWS event
- Self Paced in your AWS environment

Note that some labs, like the labs on model customization can only be run in your own (or an AWS customers') account. Furthermore the models you can access from this workshop account will be limited. Please contact your AWS workshop instructor or your AWS account team for more information.

Previous Next

Sandbox - Clair

Amazon Bedrock Workshop
Prerequisites
At an AWS Event (Setup)
Self paced (Setup)
Prompt Engineering
Text Generation
Knowledge Bases and RAG
Model Customization
Image and Video Generation Applications
Agents
Open Source With Bedrock

AWS account access
Open AWS console (us-west-2)
Get AWS CLI credentials

Content preferences
Language
English

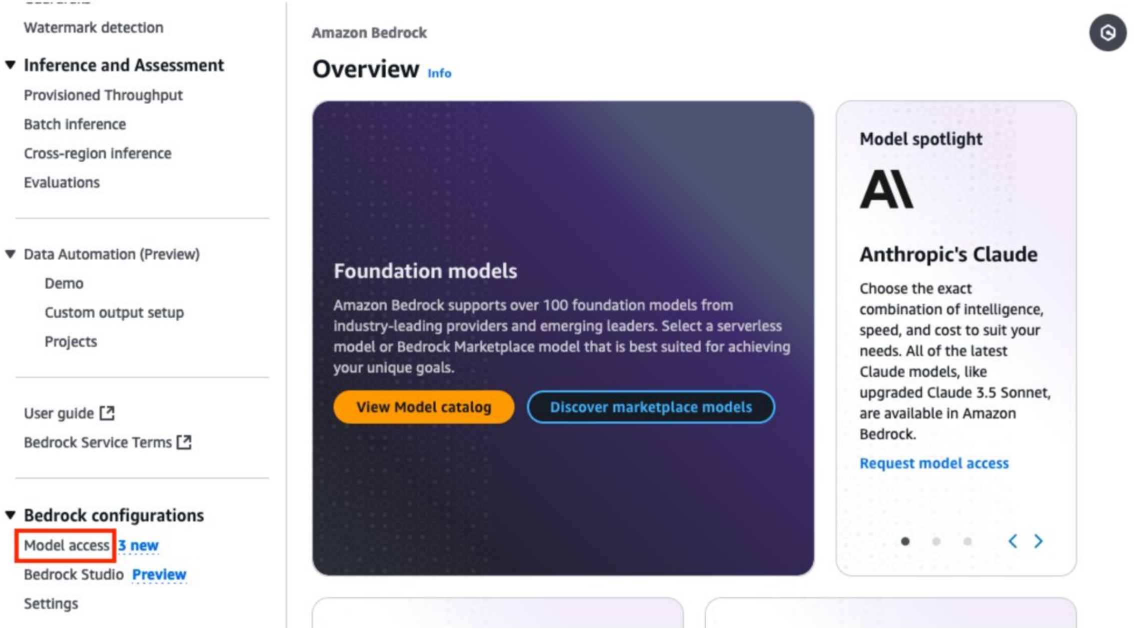
Exit event

Event ends in 2 days 4 hours 48 minutes.

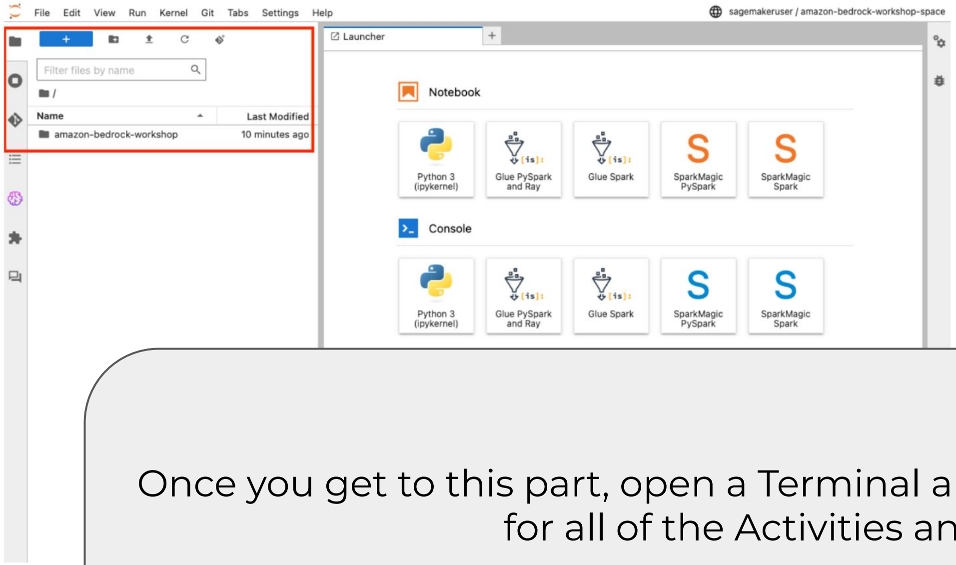
At an AWS Event (Setup)

If you are running this workshop in an AWS event, an AWS account will be provisioned for you.

To get started, go to AWS Console. Go to Amazon Bedrock console and on left menu, click on **Model access**:



- Click on **File Browser** to view already cloned git repository and to launch available notebooks:



Once you get to this part, open a Terminal and clone the workshop repository for all of the Activities and Challenges:

git clone https://github.com/ClairSullivan-Associates/genai_foundations.git

Important
Make sure the source code has been d

- When you open your first notebook,

Download code sample repository

If you cannot see the source code in SageMaker Studio (JupyterLab), cl … the following command in a Terminal window in SageMaker Studio (JupyterLab):

```
git clone https://github.com/aws-samples/amazon-bedrock-workshop
```

How to use SageMaker notebooks

A quick note on throttling

```
ERROR:root:Error raised by bedrock service: An error
occurred (ThrottlingException) when calling the InvokeModel
operation (reached max retries: 4): Too many tokens, please
wait before trying again.
```

Some Notes for Module 1

- Don’t worry (too much) about rewriting the prompts
  - Just try to ask good questions through the Human Prompt
- Some problems might be easy to solve with simple coding, but let’s stick to making the LLMs do it for us
- All code, slides can be found on the repo

https://github.com/ClairSullivan-Associates/genai_foundations

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